Final runs — cross-dataset pruning (mean ± std over 3 seeds, final-epoch checkpoint)

Metrics: Acc = accuracy, BalAcc = balanced accuracy, mF1 = macro-F1, AUC.
lr = fine-tune learning rate. Dense = unpruned baseline.

FloodNet (optical UAV)

Arch	Method	Ratio	lr	Acc	BalAcc	macro-F1	AUC
ResNet18	dense	0%	0.01	0.970±0.004	0.914±0.016	0.929±0.011	0.991±0.003
ResNet18	unstructured	50%	0.01	0.965±0.004	0.900±0.010	0.917±0.010	0.995±0.002
ResNet18	unstructured	90%	0.01	0.960±0.004	0.920±0.010	0.911±0.009	0.989±0.002
ResNet18	structured	30%	0.01	0.968±0.005	0.913±0.016	0.925±0.012	0.986±0.012
ResNet18	structured	50%	0.01	0.970±0.007	0.909±0.011	0.928±0.016	0.989±0.004
ResNet18	structured	70%	0.01	0.938±0.026	0.856±0.076	0.857±0.063	0.930±0.052
ResNet18	structured	90%	0.01	0.893±0.059	0.836±0.047	0.798±0.064	0.930±0.018
ResNet18	structured	50%	0.001	0.960±0.005	0.943±0.003	0.914±0.009	0.981±0.003
ResNet18	structured	90%	0.001	0.807±0.093	0.778±0.045	0.699±0.093	0.881±0.051
VGG-16	dense	0%	0.01	0.967±0.002	0.924±0.009	0.924±0.006	0.990±0.000
VGG-16	unstructured	50%	0.01	0.967±0.002	0.924±0.009	0.924±0.006	0.990±0.001
VGG-16	unstructured	90%	0.01	0.975±0.007	0.940±0.023	0.943±0.015	0.980±0.020
VGG-16	structured	30%	0.01	0.603±0.339	0.528±0.039	0.386±0.199	0.737±0.035
VGG-16	structured	50%	0.01	0.883±0.058	0.887±0.032	0.803±0.076	0.940±0.027
VGG-16	structured	70%	0.01	0.812±0.041	0.664±0.124	0.621±0.064	0.675±0.187
VGG-16	structured	90%	0.01	0.375±0.353	0.500±0.000	0.230±0.167	0.500±0.000
VGG-16	structured	70%	0.001	0.500±0.374	0.500±0.000	0.289±0.177	0.922±0.003

SEN1FLOODS11 (Sentinel-1 SAR)

Arch	Method	Ratio	lr	Acc	BalAcc	macro-F1	AUC
ResNet18	dense	0%	0.01	0.840±0.009	0.840±0.007	0.827±0.009	0.908±0.013
ResNet18	unstructured	50%	0.01	0.845±0.006	0.850±0.007	0.833±0.006	0.912±0.011
ResNet18	unstructured	90%	0.01	0.815±0.013	0.811±0.013	0.799±0.013	0.888±0.017
ResNet18	structured	30%	0.01	0.783±0.032	0.786±0.026	0.768±0.030	0.858±0.015
ResNet18	structured	50%	0.01	0.780±0.006	0.767±0.008	0.759±0.006	0.841±0.010
ResNet18	structured	70%	0.01	0.748±0.033	0.758±0.020	0.734±0.028	0.834±0.013
ResNet18	structured	90%	0.01	0.578±0.038	0.656±0.036	0.577±0.038	0.763±0.026
ResNet18	structured	50%	0.001	0.731±0.016	0.744±0.002	0.718±0.010	0.833±0.015
ResNet18	structured	90%	0.001	0.618±0.020	0.646±0.005	0.609±0.013	0.725±0.006
VGG-16	dense	0%	0.01	0.661±0.230	0.709±0.148	0.621±0.262	0.858±0.003
VGG-16	unstructured	50%	0.01	0.641±0.217	0.691±0.135	0.601±0.248	0.849±0.005
VGG-16	unstructured	90%	0.01	0.638±0.215	0.687±0.132	0.597±0.245	0.846±0.014
VGG-16	structured	30%	0.01	0.632±0.211	0.669±0.120	0.586±0.237	0.834±0.005
VGG-16	structured	50%	0.01	0.548±0.151	0.643±0.101	0.519±0.190	0.788±0.007
VGG-16	structured	70%	0.01	0.445±0.156	0.500±0.000	0.300±0.070	0.502±0.225
VGG-16	structured	90%	0.01	0.445±0.156	0.500±0.000	0.300±0.070	0.500±0.000
VGG-16	structured	70%	0.001	0.500±0.165	0.500±0.000	0.325±0.074	0.700±0.000

One-line reading

- Unstructured pruning transfers across both modalities (ResNet18 holds to 90% on both; macro-F1 within ~0.03 of dense).
- Structured pruning degrades SAR earlier and more steadily than optical on ResNet18 (SAR drops from 30%; optical stable to 50%).
- Lower fine-tune lr (0.001) does NOT rescue the SAR structured drop the drop is not a fine-tuning-recipe artifact.

- VGG-16 is unstable across seeds (dense SAR 0.66±0.23; structured collapses on BOTH datasets) VGG-16 structured pruning is unreliable regardless of sensor.